Computer Networks

Introduction to Computer Networks

- Concepts & Definitions
- History & Evolution
- Uses & Applications
- Advantages & Disadvantages
- Packet Tracer Lab

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Session Roadmap

01

Why Networks?

The problem networking solves

02

Concepts & Definitions

What a network actually is

03

History & Evolution

ARPANET to the modern Internet

04

Uses & Applications

Networking in the real world

05

Advantages & Risks

The balanced view

06

Practical: Packet Tracer

Build your first network

LESSON

Weekly Learning Outcomes

By the end of this session you will be able to:

01

Define a Computer Network

Explain what a network is, name its parts, and say why organisations build them.

02

Trace Its History

Identify the milestones from ARPANET (1969) through TCP/IP and the Web to today.

03

Identify Real Applications

Recognise resource sharing, communication, e-commerce, cloud, and IoT in daily life.

04

Weigh Benefits & Risks

Discuss what networking gives an organisation, and what it exposes it to.

01

Why Do We Need Networks?

Before we define a network, let us understand the problem it solves

- The standalone computer
- Sneakernet
- Why we communicate
- Business drivers

The Standalone Computer Problem

Imagine an office in 1980: every computer is an island

A computer that is not connected to anything can only use what is physically inside it or plugged into it. Its data cannot leave, and no other machine's data can get in. Everything it knows, it knows alone.

One printer per person

Every desk needs its own printer. Buy 50 computers, buy 50 printers.

Data cannot move

The report on your machine is trapped there until you physically carry it.

No shared truth

Ten people keep ten copies of the price list. Which one is correct?

Everything is slow

Sending a file to the next building means walking there and back.

Backups are manual

If that one hard disk fails, the work is simply gone.

No collaboration

Two people cannot work on the same document at the same time.

The core issue: computing power is useless in isolation. Value appears when machines can share what they have and what they know.

"Sneakernet": How We Shared Before

Copy to a disk, walk it across the office, copy it back — named after the sneakers you wore



Total time to move one file across the office: 10–30 minutes · Capacity of one floppy disk: 1.44 MB (less than one photo today)

What sneakernet teaches us

Slow

Humans move at walking speed. Data should not.

Error-prone

Which copy is the latest? Everyone has a different one.

Does not scale

Fine for 2 people. Impossible for 200.

The realisation: if we connect the machines directly, the file moves in under a second — and there is only ever one copy. That is a network.

Why Do We Need to Communicate?

Networking is not really about cables — it is about the human need to share

Long before computers, people built systems to move information across distance: messengers, postal services, the telegraph, the telephone. Each one existed because separated people needed to act on the same information. Computer networks are simply the newest answer to a very old need.

To share what we know

One person learns something useful; everyone else benefits only if it travels.

To coordinate

A team can only act as one if all members hold the same picture of reality.

To decide faster

A decision is only as fresh as the information behind it. Delay costs money.

To cross distance

Geography stops mattering once information can travel faster than people.

Key idea: a computer network is a communication system. Everything else in this course — cables, protocols, routers — exists to serve that one purpose.

Why Do Organisations Build Networks?

Four business drivers that justify the cost every single time

1

Save Money

Twenty staff sharing one network printer and one licence pool costs far less than twenty of everything.

One ₨ 200,000 printer instead of 20 × ₨ 15,000

2

One Version of the Truth

A shared database means every department reads and writes the same record, so no one works from stale data.

Student records: one system, all faculties

3

Speed of Work

Information reaches the person who needs it the moment it exists, instead of the next day.

Exam results published instantly, not posted

4

Room to Grow

Adding a new employee means plugging in one more device, not rebuilding the system.

New lab of 40 PCs joins the existing campus LAN

Case Study: A University Campus

The same institution, with and without a network

Everyday task	Without a network	With a network
Registering for a course	Queue at the office, fill a paper form, wait days	Log in from your phone, confirmed in seconds
Getting lecture notes	Photocopy from a friend, hope it is the latest	Download from the LMS, always current
Checking library books	Walk to the library and search the shelves	Search the online catalogue from anywhere
Publishing exam results	Print, pin to a noticeboard, students travel in	Published once, every student sees it at once
Backing up research	Copy to a personal drive and hope	Automatic nightly backup to a central server

Notice: the network did not change what the university does. It changed how fast, how accurately, and how far it can do it.

02

Concepts & Definitions

What a computer network is, and the parts it is built from

- Definition
- Nodes, Links, Protocols
- Key terminology
- Packets
- Client-server vs P2P
- Network sizes

What Is a Computer Network?

The formal definition, unpacked word by word

> *"A computer network is a collection of two or more autonomous computing devices, interconnected by a transmission medium, that exchange data according to a shared set of rules."*

Unpacking the definition

Two or more	One device alone is not a network. Communication needs at least two parties.
Autonomous	Each device works on its own. None of them controls or depends on another to function.
Interconnected	There is a real physical path between them: copper, fiber, or a radio signal.
Exchange data	The whole point. Information flows in both directions, not just one.
Shared rules	Both sides agree in advance on the format and timing. We call these rules protocols.

Every Network Has Three Ingredients

Remove any one of them and communication stops



1. Nodes

The devices that send, receive, or forward data.

PCs, phones, servers, printers, routers

2. Links

The physical path the signal actually travels along.

Copper cable, optical fiber, Wi-Fi, 4G/5G

3. Protocols

The agreed rules for format, order, and timing.

TCP/IP, HTTP, Ethernet, Wi-Fi (802.11)

Your home Wi-Fi: laptop, phone, and smart TV are the nodes · Wi-Fi radio is the link · TCP/IP is the protocol. That is a real network.

Ingredient 1: Nodes

Anything on the network with an address is a node

A node is any device that can send, receive, or pass along data. Nodes fall into two broad families: end devices, which people actually use, and intermediary devices, which exist only to move other devices' traffic along.

END DEVICES

Where data starts or finishes its journey

- **Computers & laptops**
  Create and consume most data
- **Phones & tablets**
  The most common node type today
- **Printers & scanners**
  Shared peripherals with their own address
- **Servers**
  Machines that exist to serve others

INTERMEDIARY DEVICES

Move other devices' data along the path

- **Switch**
  Connects devices inside one local network
- **Router**
  Joins separate networks and picks the route
- **Access point**
  Lets wireless devices join the wired network
- **Firewall**
  Filters traffic and enforces security rules

Ingredient 2: Links

The physical road the data travels on — wired or wireless

Wired Media

Signal travels through a solid cable

Twisted pair (Cat5e/Cat6)
Cheap, everywhere, up to ~100 m

Coaxial cable
Older; still used for cable TV/internet

Optical fiber
Light in glass; fastest, longest reach

Wireless Media

Signal travels through open air as radio waves

Wi-Fi (802.11)
Local coverage inside a building

Mobile (4G / 5G)
Wide-area coverage from cell towers

Bluetooth / Infrared
Very short range, low power

The trade-off

Wired wins on
speed, reliability, security

Wireless wins on
convenience, mobility, cost to install

Most real networks
use both together

Ingredient 3: Protocols

Cables move signals. Protocols make those signals mean something.

Think of a phone call to a stranger. You both agreed, without ever discussing it, to speak one at a time, in a shared language, starting with “hello” and ending with “goodbye”. Break any of those rules and the call fails — even though the line worked perfectly.

What every protocol must agree on

Syntax

What the data looks like: the format, the field order, how many bits each part takes.

Semantics

What the data means: which bits are the address, which are the message, which are the checks.

Timing

When to send and how fast, so a fast sender does not overwhelm a slow receiver.

Protocols you already use every day

HTTP / HTTPS
Loads every web page you open

TCP / IP
Delivers and addresses nearly all Internet traffic

Ethernet
Moves frames across a wired LAN

SMTP
Sends your email

DNS
Turns a name into an address

Essential Terminology

The vocabulary you will hear in every remaining week of this course

Data

Raw information: text, an image, a video frame.

Packet

A small chunk of data with an address attached, ready to travel.

Bandwidth

How much data a link can carry per second (Mbps, Gbps). The width of the road.

Latency

How long one packet takes to arrive (ms). The length of the road.

Throughput

How much data actually gets through in practice, which is always ≤ bandwidth.

IP address

The unique number identifying a device, e.g. 192.168.1.10.

Host

Any end device with an IP address that a person or service uses.

Topology

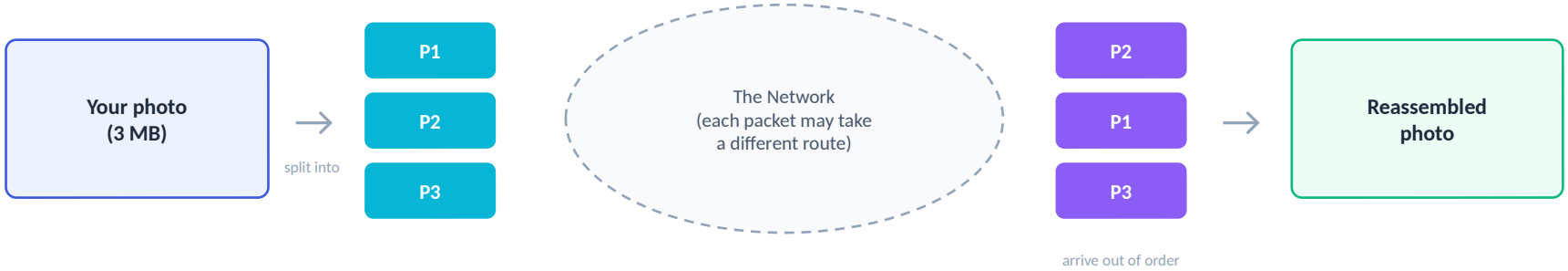
The shape of the network: how the nodes are arranged and wired.

Bandwidth vs latency: a truck full of hard drives has enormous bandwidth but terrible latency. A phone call has tiny bandwidth but instant latency.

How Data Actually Travels: Packets

Big messages are never sent whole — they are chopped up first

When you send a photo, the network does not send it as one huge block. It breaks the file into small numbered packets, sends each one separately, then reassembles them in order at the far end. If one packet is lost, only that one is re-sent.



Why packets, and not one big block?

Sharing

Many conversations can interleave on one cable at the same time.

Resilience

Lose one small packet, re-send only that — not the whole 3 MB file.

Flexible routing

If one path breaks, later packets simply go around it.

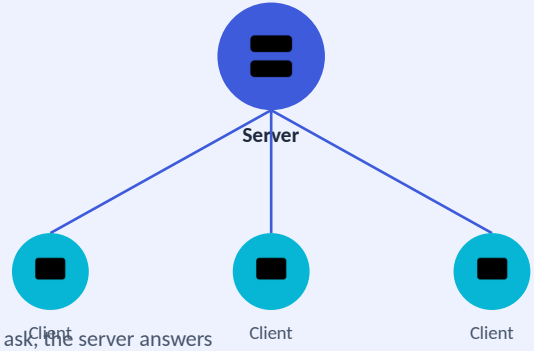
This idea has a name: packet switching. It was the breakthrough that made the Internet possible — we meet its inventor in the next section.

Two Ways to Organise a Network

Who serves whom? The answer shapes the whole design.

CLIENT–SERVER

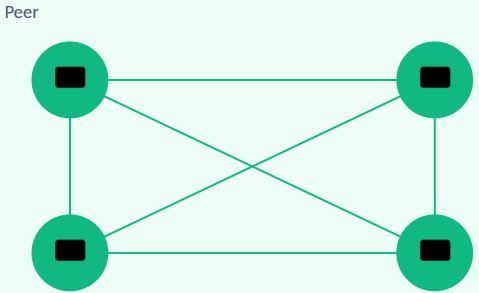
One central machine serves many



- Clients ask, the server answers
- Easy to manage and secure centrally
- If the server dies, everything stops
- Used by: web, email, this university's LMS

PEER-TO-PEER (P2P)

Every device is both client and server



- Any peer can serve any other peer
- No single point of failure, cheap to start
- Hard to secure and manage at scale
- Used by: BitTorrent, blockchain, file sharing

Networks Come in Different Sizes

Same idea, wildly different scale — a preview of Week 3

PAN

~10 metres

Personal Area Network

Your phone paired to earbuds and a smartwatch

LAN

One building or site

Local Area Network

This computer lab · your home Wi-Fi · an office floor

MAN

A city

Metropolitan Area Network

A city council linking all its branch offices across Anuradhapura

WAN

Countries, continents

Wide Area Network

The Internet itself · a bank linking Colombo to London

The key insight: the Internet is not a special kind of network. It is simply an enormous WAN made by connecting millions of smaller LANs together.

03

History & Evolution

From a four-node experiment to a network of billions of devices

Circuit switching

Packet switching

ARPANET 1969

TCP/IP 1983

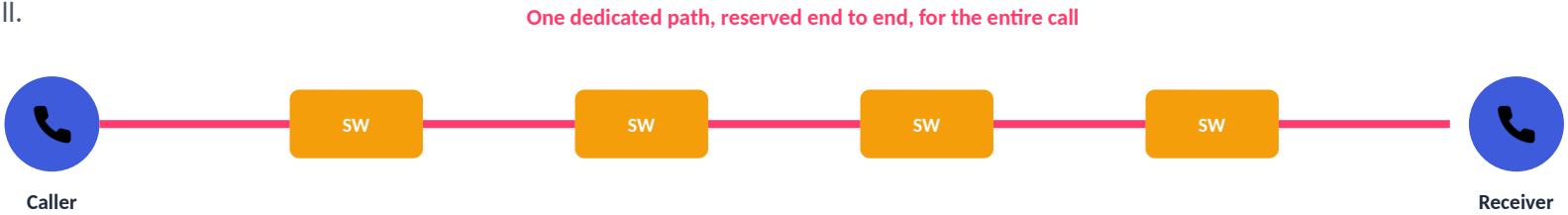
The Web 1991

Mobile, cloud, IoT

Before Computers: The Telephone Network

The world already had a network — it just was not built for data

By the 1900s the telephone network spanned the globe. It worked by circuit switching: when you called someone, an operator (later a machine) reserved a complete, unbroken electrical path from your phone to theirs. That path stayed yours, and nobody else's, for the whole call.



Why this was wrong for computers

Wasteful

Computers send data in short bursts, then sit silent. The reserved line sat idle most of the time.

Fragile

Cut the line anywhere along the path and the entire conversation drops instantly.

Does not scale

Every simultaneous conversation needs its own physical path. The maths breaks down fast.

The Big Idea: Packet Switching

Three researchers, working separately, had the same thought in the early 1960s

Paul Baran

RAND Corporation, USA

Designed a network with no central point, so it could survive losing many of its nodes.

Donald Davies

NPL, United Kingdom

Reached the same design independently, and gave us the word “packet”.

Leonard Kleinrock

MIT, later UCLA

Built the mathematical theory showing why chopping data into queued blocks works.

CIRCUIT SWITCHING

- Reserve one whole path before sending
- Path stays idle between bursts of data
- One cut kills the conversation
- Predictable, but wasteful

PACKET SWITCHING

- No reservation — just address each packet and send
- Many conversations share the same line
- A cut? Later packets route around it
- Efficient, resilient, and it scales

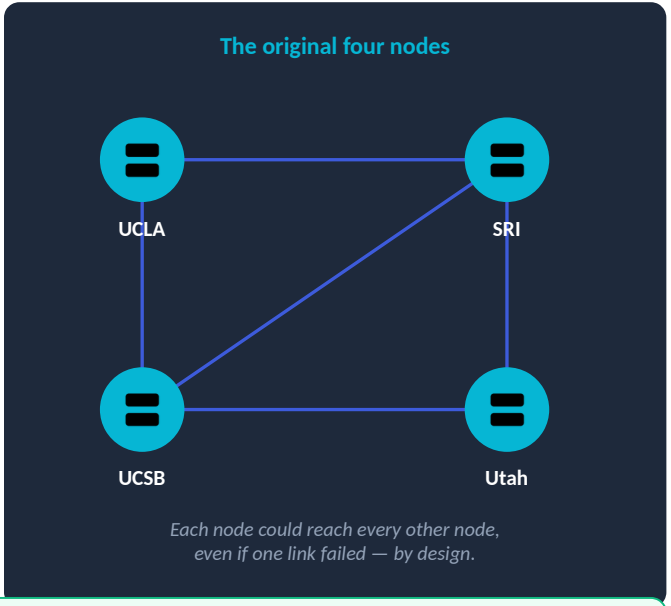
Why it mattered: packet switching is the single idea that separates the Internet from the telephone system. Every slide after this one depends on it.

1969: ARPANET Goes Live

The first packet-switched network — four computers, one very big idea

Funded by ARPA (the US defence research agency), ARPANET connected four university research computers so they could share scarce, expensive computing time. On 29 October 1969 the first message was sent from UCLA to Stanford. The team typed “LOGIN”. The system crashed after two letters, so the first word ever sent over the ancestor of the Internet was “LO”.

Nodes at launch	4
Link speed	50 kbps
First message	“LO” (it crashed)
Nodes by 1971	15



The lasting lesson: ARPANET proved packet switching worked in the real world, not just on paper. Everything since has been scaling up that proof.

The 1970s: Email and the Invention of TCP/IP

The network grows, and a new problem appears: networks that cannot talk to each other

1971

Email Is Invented

Ray Tomlinson sends the first network email and picks the @ symbol to separate user from machine. Within two years, email is 75% of all ARPANET traffic — nobody had planned for that.

1973

The Interconnection Problem

Other networks now exist (satellite, radio, other universities), but each speaks its own language. They cannot exchange data. A network of networks needs a common tongue.

1974

TCP/IP Is Designed

Vint Cerf and Bob Kahn publish a protocol that any network can adopt: IP handles addressing and routing, TCP handles reliable delivery. This is the moment “internetworking” is born.

Notice the pattern: the killer application (email) was an accident. The engineers built a machine for sharing computers; people used it to talk to each other.

1 January 1983: The Internet Is Born

The day every machine on ARPANET had to switch protocol — or go silent

“Flag Day”: on one fixed date, every host on ARPANET stopped using the old NCP protocol and switched to TCP/IP. There was no gradual migration and no fallback. Operators wore badges reading “I survived the TCP/IP transition”.

Why 1983 is the Internet's real birthday

One common language

Any network anywhere could now join simply by speaking TCP/IP. No permission needed.

A network of networks

“Internet” literally means inter-network. From this date, that is what existed.

DNS arrives (1983–85)

Names replace numbers. You type a name; DNS finds the address behind it.

It still runs today

The TCP/IP you will configure in the lab this week is the same suite adopted in 1983.

1991: The World Wide Web

The Internet existed for years before the Web. They are not the same thing.

The Internet

The infrastructure. Cables, routers, and TCP/IP that move packets between machines. Built from 1969 onward.

The Web

One application that runs on top of the Internet: linked documents you view in a browser. Invented in 1991.

Tim Berners-Lee, at CERN, gave the Web three pieces

HTML

A language for writing documents with links inside them.

HTTP

A protocol for asking a server for a document and receiving it.

URL

A universal address so any document anywhere can be named and found.

The decision that changed everything: CERN released the Web into the public domain in 1993. No licence, no royalties. Anyone could build on it — and everyone did.

1990s–2000s: The Internet Goes Public

From a researchers' tool to household infrastructure in about fifteen years

1990s

Dial-Up and the Dot-Com Boom

Home users connect over telephone lines at 56 kbps. Browsers, ISPs, and the first online businesses appear. Everyone rushes online; many companies fail in the 2000 crash, but the infrastructure they built stays.

- 56 kbps modems
- First e-commerce
- Search engines

2000s

Broadband Changes the Rules

ADSL and fiber make connections always-on and hundreds of times faster. Once you no longer pay by the minute, whole new applications become possible: streaming video, social media, online gaming.

- Always-on
- Streaming
- Social media

2007+

Mobile and Cloud

The smartphone puts a networked computer in every pocket. At the same time, cloud platforms let anyone rent computing power by the hour instead of buying servers. Access becomes constant; capacity becomes elastic.

- Smartphones
- 4G
- AWS / cloud

2010s–Today: Everything Connects

The network stops being something you log into and becomes something you live inside

For fifty years, networks connected computers that people operated. The shift now underway is that most connected devices have no user at all. Sensors, meters, vehicles, and machines exchange data with each other automatically. Devices now vastly outnumber people online.

Internet of Things

Everyday objects with sensors and network connections, reporting and reacting on their own.

5G

Very high speed and very low delay, enough for real-time control of remote machinery.

Cloud & Edge

Computing moves off your device entirely — or right next to it, wherever it is cheapest.

AI Workloads

Training and serving models moves enormous datasets between data centres constantly.

~5.5 bn

people online worldwide

3–4×

more devices than people online

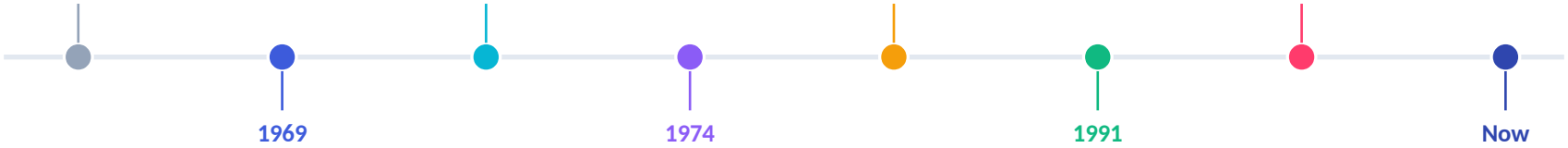
< 10 ms

typical 5G round-trip delay

Figures are approximate and change constantly — quote them as orders of magnitude, not exact values.

The Complete Timeline

Fifty-five years, at a glance — this is the sequence to remember



1960s

Packet Switching

Baran, Davies, and Kleinrock design a network with no centre.

1969

ARPANET

Four university nodes exchange the first packets.

1971

Email

Tomlinson sends the first network mail and picks @.

1974

TCP/IP Designed

Cerf and Kahn publish the protocol that unites all networks.

1983

Internet Born

Flag Day: everyone switches to TCP/IP. DNS follows.

1991

World Wide Web

Berners-Lee releases HTML, HTTP, and URLs.

2000s

Broadband & Mobile

Always-on links; the smartphone arrives in 2007.

Now

IoT, 5G & Cloud

Billions of unattended devices, always connected.

04

Uses & Applications

Where all of this theory shows up in real organisations and real life

- Resource sharing
- Communication
- E-commerce
- Cloud computing
- IoT

Application 1: Resource Sharing

The original reason networks were built, and still the most common

A resource is anything useful that costs money: a printer, a disk, a database, a software licence, or processing power. Without a network, each user needs their own copy of everything. With a network, one resource serves everybody.

Hardware sharing

One printer, one scanner, one high-capacity storage array used by an entire department.

Data sharing

A single database that every user reads from and writes to, so there is one correct version.

Software sharing

Applications and licences installed centrally, updated once, and delivered to every desk.

Processing sharing

Heavy computation offloaded to a powerful shared server rather than done on each PC.

You already use: Campus print server · Google Drive & Dropbox · shared network drives · university library database · the HPC cluster used by researchers

In practice: This computer lab: 40 PCs, but one printer, one file server, and one licence pool. Buying 40 of each would cost roughly 40 times more.

Application 2: Communication

The application nobody planned for, that became the biggest of all

Networks let people exchange information instantly regardless of distance. This started as plain text email in 1971 and now includes voice, video, and real-time collaboration. It is, by traffic and by hours used, what most people actually do with networks.

Asynchronous

You send, they read later: email, forums, comments. No one has to be present at the same time.

Real-time text

Instant messaging and group chat: WhatsApp, Slack, Teams. Fast, informal, always on.

Voice & video

Calls carried as data packets instead of over the phone network: Zoom, Meet, VoIP.

Collaboration

Several people editing one document at the same time, each seeing the others' changes live.

You already use: Email (SMTP/POP) · WhatsApp & Viber · Zoom & Microsoft Teams · Google Docs · online lectures · social media

In practice: An online lecture: your voice becomes packets, crosses several networks, and is reassembled in 60 students' headphones in under a fifth of a second.

Application 3: E-Commerce

When networks became trustworthy enough to carry money

E-commerce is buying and selling conducted over a network. It only became possible once encryption made it safe to send payment details across a shared, public medium. It reshaped retail by removing the need for a shop to be physically near its customer.

B2C — business to consumer

A company sells directly to the public: Amazon, Daraz, an airline's booking site.

B2B — business to business

Companies buy from each other: suppliers, wholesalers, procurement systems.

C2C — consumer to consumer

People sell to each other through a platform: ikman.lk, eBay, marketplace apps.

Digital payments

The money itself moves as data: online banking, card payments, mobile wallets.

You already use: Amazon & Alibaba · Daraz & ikman.lk · PayPal & Stripe · online banking · PickMe & Uber · airline booking

In practice: Buying one book online touches several networks: the shop's web servers, a payment gateway, your bank, and a courier's tracking system — in a few seconds.

Application 4: Cloud Computing

Stop buying computers. Rent them by the hour, over the network.

Cloud computing means the storage and processing happen on someone else's machines in a data centre, and reach you over the network. You pay for what you use. The network is what makes a distant computer feel like a local one.

IaaS — infrastructure

Rent raw virtual machines, storage, and networking. You manage everything above that.

PaaS — platform

Rent a ready environment to run your code. The provider handles the servers underneath.

SaaS — software

Just use the finished application in a browser. You manage nothing: Gmail, Microsoft 365.

Storage & backup

Files kept off-site, replicated automatically, reachable from any device you log in from.

You already use: AWS, Microsoft Azure, Google Cloud · Google Drive & OneDrive · Netflix & Spotify · Microsoft 365 · GitHub

In practice: Netflix owns no cinema and ships no discs. It stores films in data centres and streams them to you as packets, on demand, wherever you are.

Application 5: Internet of Things (IoT)

Networks for machines, not people

IoT means giving ordinary physical objects a sensor, a small processor, and a network connection so they can report what they observe and act on what they receive. There is no user at the keyboard. The devices talk to each other and to servers, continuously.

Sense

A device measures something real: temperature, motion, location, soil moisture, heart rate.

Send

It transmits that reading over Wi-Fi, 4G/5G, or a low-power radio link to a server.

Analyse

Software combines readings from many devices and decides what, if anything, should happen.

Act

A command goes back out: switch the pump on, dim the light, apply the brakes, raise an alert.

You already use: Smart homes · fitness watches · smart electricity meters · precision agriculture sensors · traffic management · industrial monitoring

In practice: Smart agriculture in Sri Lanka: soil sensors report moisture over a mobile network; a server decides when to irrigate and switches the pump on automatically.

A Day With Networks

All five applications, in one ordinary day, without you noticing any of them



07:00

Wi-Fi & Email

Your phone rejoins the home Wi-Fi and pulls down overnight mail.

Communication



10:00

Shared Lab PC

You log into a lab machine and print to the shared department printer.

Resource sharing



13:00

Order Lunch

An app takes your order and your payment across three separate networks.

E-commerce



20:00

Streaming

A film reaches your screen from a data centre thousands of km away.

Cloud



23:00

Smart Home

Lights and thermostat exchange readings and settle for the night.

IoT

The point: a well-built network is invisible. You only ever notice it on the day it stops working.

05

Advantages & Disadvantages

Every network is a trade: real benefits bought with real risks

The benefits

The risks

Managing the trade-off

Advantages of Networking

Why every organisation, despite the risks, builds one anyway

Lower Cost

Sharing hardware, licences, and connections spreads one purchase across many users.

One Rs 200,000 printer serves 40 staff instead of 40 separate printers.

Centralised Data

One authoritative copy that everyone reads and writes, backed up in one place.

Student records: one database, every faculty sees the same result.

Speed

Information arrives the instant it exists, not the next working day.

Exam results reach every student at the same moment, instantly.

Scalability

Growth means plugging in one more device, not redesigning the system.

A new 40-PC lab simply joins the existing campus LAN.

Remote Access

Work is tied to an account, not to a desk. Location stops mattering.

Staff mark assignments from home over a VPN.

Reliability & Backup

Automatic replication means one failed disk is an inconvenience, not a disaster.

Nightly backup to a central server, with no human action needed.

Disadvantages and Risks of Networking

The same connectivity that delivers the benefits also creates the exposure

Security Risks

A connected machine can be reached by anyone else who is also connected — including attackers.

Malware, ransomware, data breaches, phishing.

Cost to Build & Run

Cables, switches, servers, licences, and skilled staff are all recurring expenses.

A campus network needs a permanent IT team.

Downtime Affects Everyone

Centralising creates single points of failure. One dead switch can stop a whole floor.

Server outage: nobody can register for anything.

Complexity

As a network grows, designing, configuring, and troubleshooting it gets rapidly harder.

One wrong subnet mask can take a day to find.

Privacy Concerns

Traffic and activity can be logged, monitored, and correlated — by the organisation or by intruders.

Browsing history, location, and message metadata.

Dependence

Once work depends on the network, no network means no work at all.

Internet drops — an online exam simply stops.

The Balanced View

We do not avoid the risks by avoiding networks. We manage them.

No organisation looks at the risk list and decides to go back to sneakernet. The benefits are too large. Instead, every risk on the previous slide has a well-understood countermeasure — and most of this course is really about those countermeasures.

Security risks	→	Firewalls, encryption, authentication, network segmentation
Downtime	→	Redundant links, backup power, failover devices, spare hardware
Complexity	→	Layered design (OSI/TCP-IP), documentation, standard protocols
Privacy	→	Access control, HTTPS everywhere, VPNs, data protection policy
Cost	→	The savings from sharing outweigh the spend — that is the whole business case

06

Practical Activity

Orientation to Cisco Packet Tracer — build your first working network

What is Packet Tracer | The interface | Build | Address | Test | Simulate

What Is Cisco Packet Tracer?

A full network laboratory that runs inside your laptop

Packet Tracer is a network simulator made by Cisco. You drag real device models onto a canvas, cable them together, configure them exactly as you would configure the physical hardware, and then watch the traffic move.

It is free

No cost to you. A real router and switch would cost tens of thousands of rupees each.

Nothing can break

Misconfigure everything, delete it, start again. No hardware is harmed, ever.

You can see the packets

Simulation mode slows time down so you can watch a single packet cross the network.

It is the industry tool

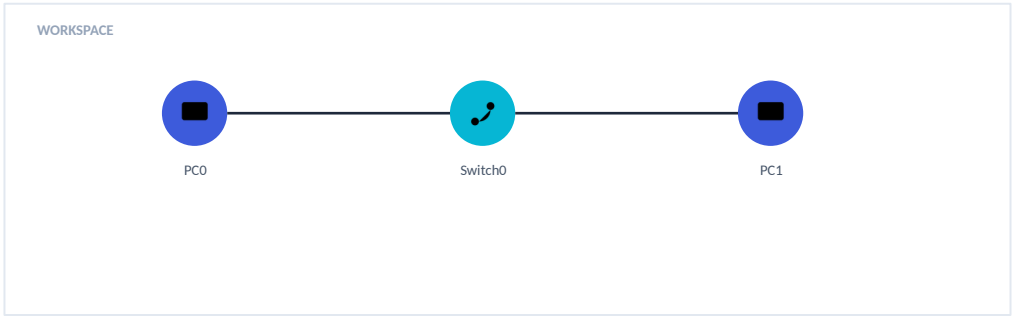
The same simulator used in Cisco's CCNA certification worldwide.

Today's session in four tasks

1. Build a two-PC network
2. Assign IP addresses
3. Test it with ping
4. Watch it in Simulation mode

The Packet Tracer Interface

Five areas do ninety percent of the work — learn these and you can build anything



1 **Workspace**

The canvas. Drag devices here and cable them together.

2 **Device Panel**

Bottom-left. Pick a category, then drag the model you want.

3 **Connections**

The lightning-bolt category. Copper straight-through is your default.

4 **Realtime / Simulation**

Top-right toggle. Realtime acts like a live network; Simulation freezes time.

5 **Device Config**

Click any device to open its Config, Desktop, and CLI tabs.

Before you start: launch Packet Tracer now and find all five areas on your own screen. Ask if any of them are missing.

Task 1

Build a Two-PC Network

Three devices, two cables, five minutes

1. From the Device Panel, choose End Devices and drag two PCs onto the workspace.
2. Choose Network Devices → Switches and drag one 2960 switch onto the workspace.
3. Choose Connections (the lightning bolt) and pick Copper Straight-Through.
4. Click PC0, choose FastEthernet0, then click Switch0 and choose FastEthernet0/1.
5. Repeat for PC1, connecting it to Switch0 port FastEthernet0/2.
6. Wait for both link lights to turn green. Amber means the port is still initialising.

Success looks like

Both cable ends show a solid green triangle.

If it goes wrong

If a light stays red, you used the wrong cable type. Delete it and use Copper Straight-Through.

Task 2

Assign IP Addresses

Without an address, a device cannot be reached — cables alone are not enough

1. Click PC0 to open its window, then select the Desktop tab.
2. Click IP Configuration and choose the Static option.
3. Set PC0's IPv4 Address to 192.168.1.10 and Subnet Mask to 255.255.255.0.
4. Close the window. Click PC1 and open Desktop → IP Configuration.
5. Set PC1's IPv4 Address to 192.168.1.11 with the same subnet mask.
6. Leave the Default Gateway blank — both PCs are on the same network, so no router is needed yet.

Success looks like

Both PCs have addresses starting 192.168.1. and the same mask.

If it goes wrong

The addresses must differ, but the subnet must match. 192.168.1.10 and 192.168.2.11 will not talk.

Task 3

Test It With Ping

Ping is the first thing every network engineer reaches for

1. Click PC0, open the Desktop tab, and choose Command Prompt.
2. Type: ping 192.168.1.11 and press Enter.
3. The first request may time out while ARP discovers PC1's MAC address. This is normal.
4. Look for four Reply from 192.168.1.11 lines with a time in milliseconds.
5. Check the summary: Packets Sent = 4, Received = 4, Lost = 0.
6. Now ping in the opposite direction, from PC1 to 192.168.1.10, to confirm both ways work.

Success looks like

Reply from 192.168.1.11 · 0% loss.

If it goes wrong

Request timed out every time? Re-check the IP addresses, the mask, and that both link lights are green.

Task 4

Watch It in Simulation Mode

Where Packet Tracer stops being a drawing tool and starts being a teaching tool

1. Switch the toggle at the bottom-right from Realtime to Simulation.
2. Click Edit Filters and show only ICMP and ARP so the view stays readable.
3. Use the Add Simple PDU tool (the closed envelope), click PC0, then click PC1.
4. Press Capture / Forward repeatedly and watch the envelope move one hop at a time.
5. Click on any envelope in the Event List to open it and inspect its layers.
6. Note the ARP exchange happening first, before the ICMP ping can be delivered.

Success looks like

You can describe, in order, what happened between the click and the reply.

If it goes wrong

This is the single most useful feature in Packet Tracer. You will use it in every week from here on.

Before You Leave the Lab

What you should be able to do, and how this connects to the assignments

Self-check: can you do all of these without notes?

- Place devices on the workspace and cable them with the correct cable type
- Assign a static IP address and subnet mask to a PC
- Run ping from a PC's command prompt and read the result correctly
- Switch to Simulation mode and follow one packet across the topology
- Save your work as a .pkt file

Save your file

File → Save As → week1_yourname.pkt. Keep every .pkt file you make this semester.

Why this matters

Assignment 1 (due Week 8) asks you to design, build, and verify a network. It starts exactly here.

Summary — Key Takeaways

Everything we covered in Week 1

Why networks exist	Isolated computers waste resources and cannot share. Networking solves a communication problem, not a hardware problem.
What a network is	Two or more autonomous devices, joined by a link, exchanging data under shared rules: nodes, links, protocols.
How it evolved	Packet switching → ARPANET (1969) → TCP/IP (1983) → the Web (1991) → broadband and mobile → cloud and IoT.
What we use it for	Resource sharing, communication, e-commerce, cloud computing, and the Internet of Things.
What we gain	Lower cost, one version of the truth, speed, scalability, remote access, and reliable backup.
What we risk	Security exposure, cost, downtime, complexity, privacy loss, and dependence — all manageable, none free.

Coming Up in Week 2

Network Devices and Physical Structures

This week we said “nodes and links” and moved on. Next week we open those up properly: what each device actually does, and how the shape you wire them into changes everything about cost, speed, and resilience.

Hardware Components

NIC, hub, switch, router, repeater, modem, and access point — what each one is really for.

Software Components

Drivers, protocol stacks, and the network services an operating system provides.

Types of Connection

Point-to-point versus multipoint, and when each one makes sense.

Physical Topology

Bus, star, ring, mesh, tree, and hybrid — compared on cost and fault tolerance.

To prepare

Review

Today's three ingredients: nodes, links, protocols

Practise

Rebuild today's two-PC network from memory, without the slides

Look ahead

Assignment 1 is issued in Week 3 and builds directly on this

Thank You

Any Questions?